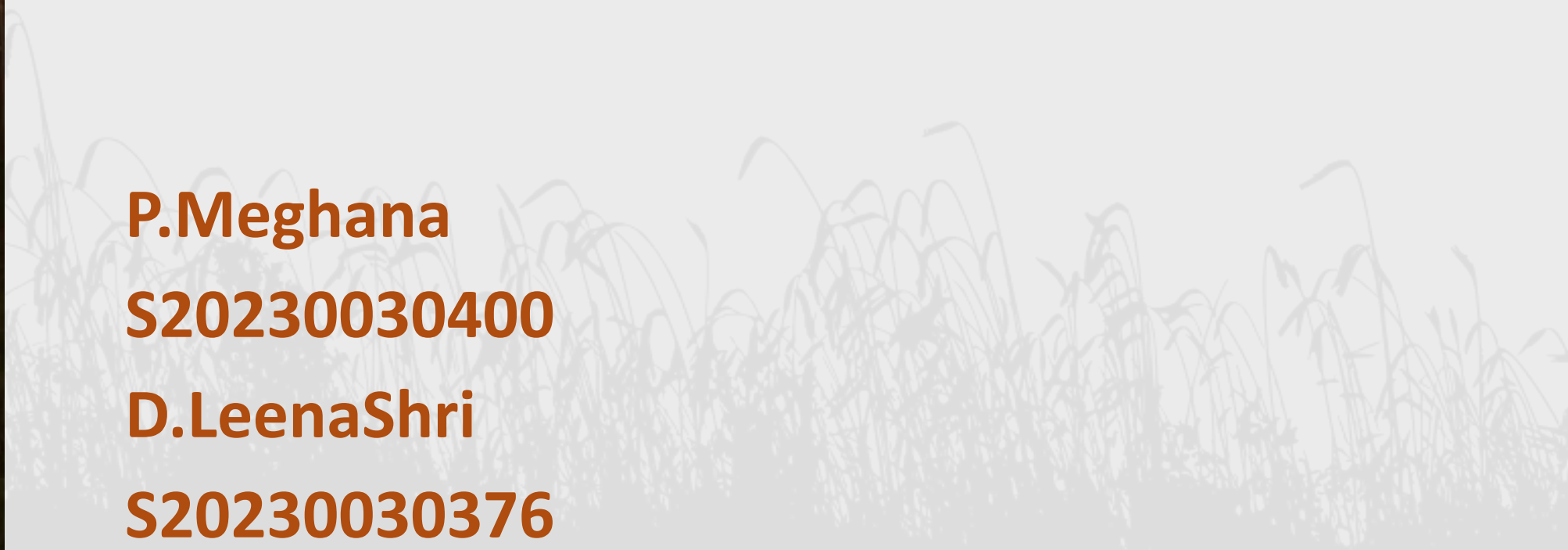




A Data-Driven Study of Forest Fire Prediction Using Exploratory Analysis and Machine Learning



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INTRODUCTION

- **Forest fires are destructive natural disasters affecting ecosystems, climate, and human life.**
- **Early prediction helps reduce damage and improves response strategies.**
- **Fire occurrence depends on:**
 - 1)Temperature**
 - 2)Relative Humidity**
 - 3)Wind Speed**
 - 4)Fuel Dryness**
- **The Major issue here is highly correlated features lead to unstable predictions.**





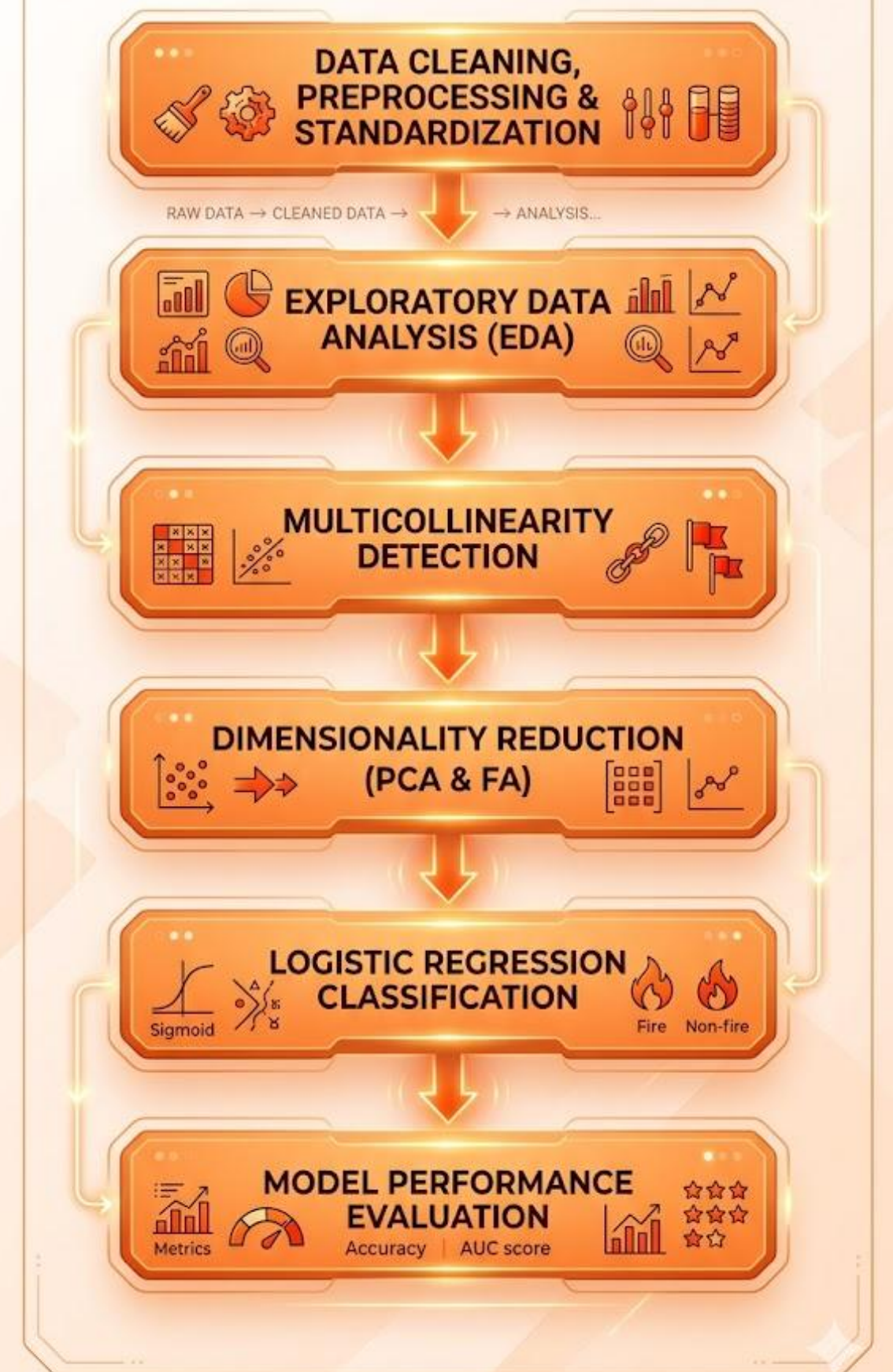
DATASET OVERVIEW

- The regions in the dataset are Bejaia and Sidi Bel-Abbes
- Time Period
 - June – September (Peak Fire Season)
- Weather Features
 - 1)Temperature
 - 2)Relative Humidity
 - 3)Wind Speed
 - 4)Rain
- Fire Indices (FWI)
 - 1)FFMC
 - 2)DMC
 - 3)DC
 - 4)ISI
 - 5)BUI
 - 6)FWI
- Target Variable
 - Fire (1) and Not Fire (0)

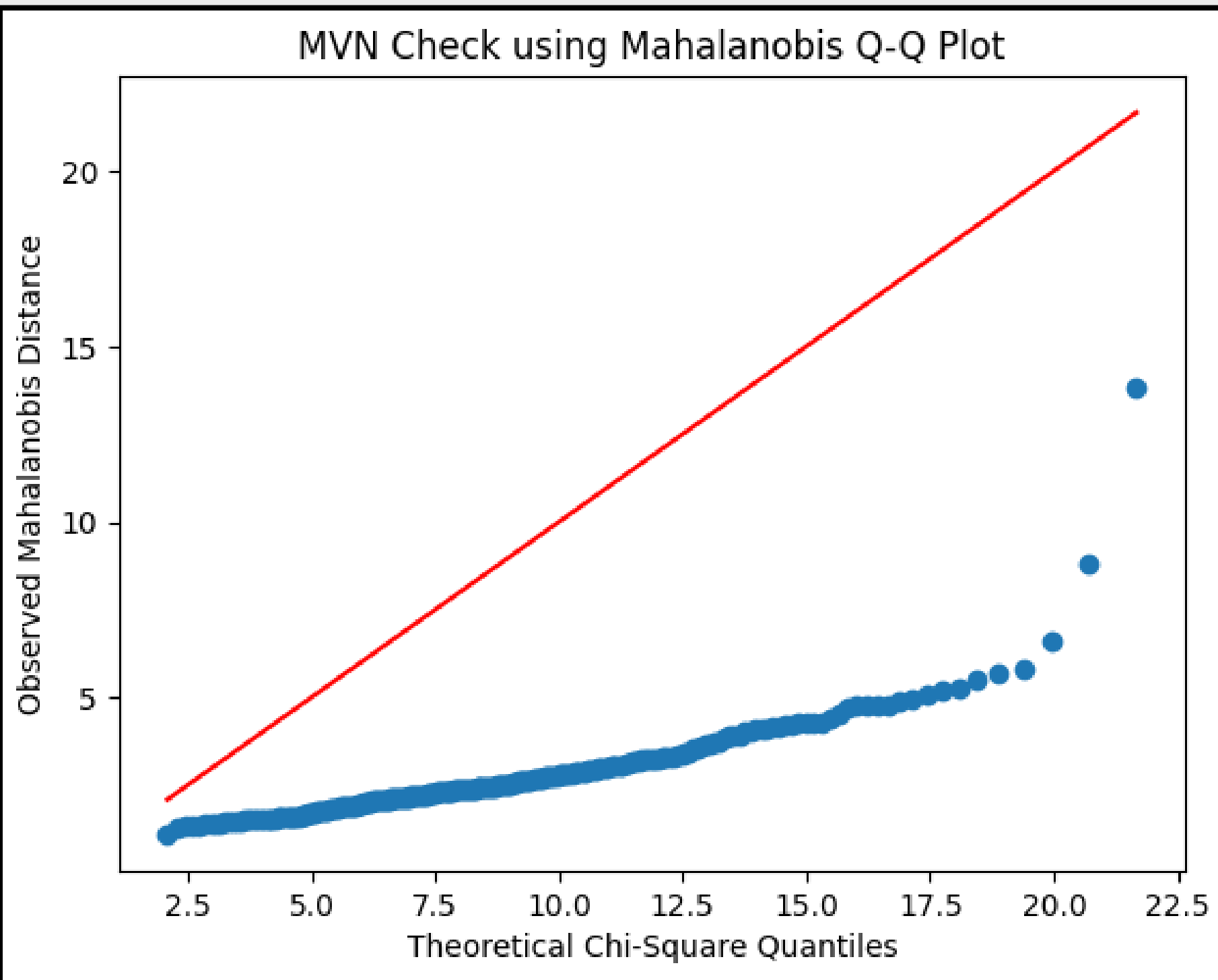
METHODOLOGY

- Data is cleaned, preprocessed, and standardized for consistency
- Exploratory Data Analysis is performed to identify patterns and relationships
- Multicollinearity among fire indices is detected using correlation analysis
- Dimensionality reduction is applied using PCA and Factor Analysis
- Logistic Regression is used to classify fire and non-fire cases
- Model performance is evaluated using Accuracy and AUC across Raw, PCA, and FA models

DATA ANALYSIS AND CLASSIFICATION WORKFLOW



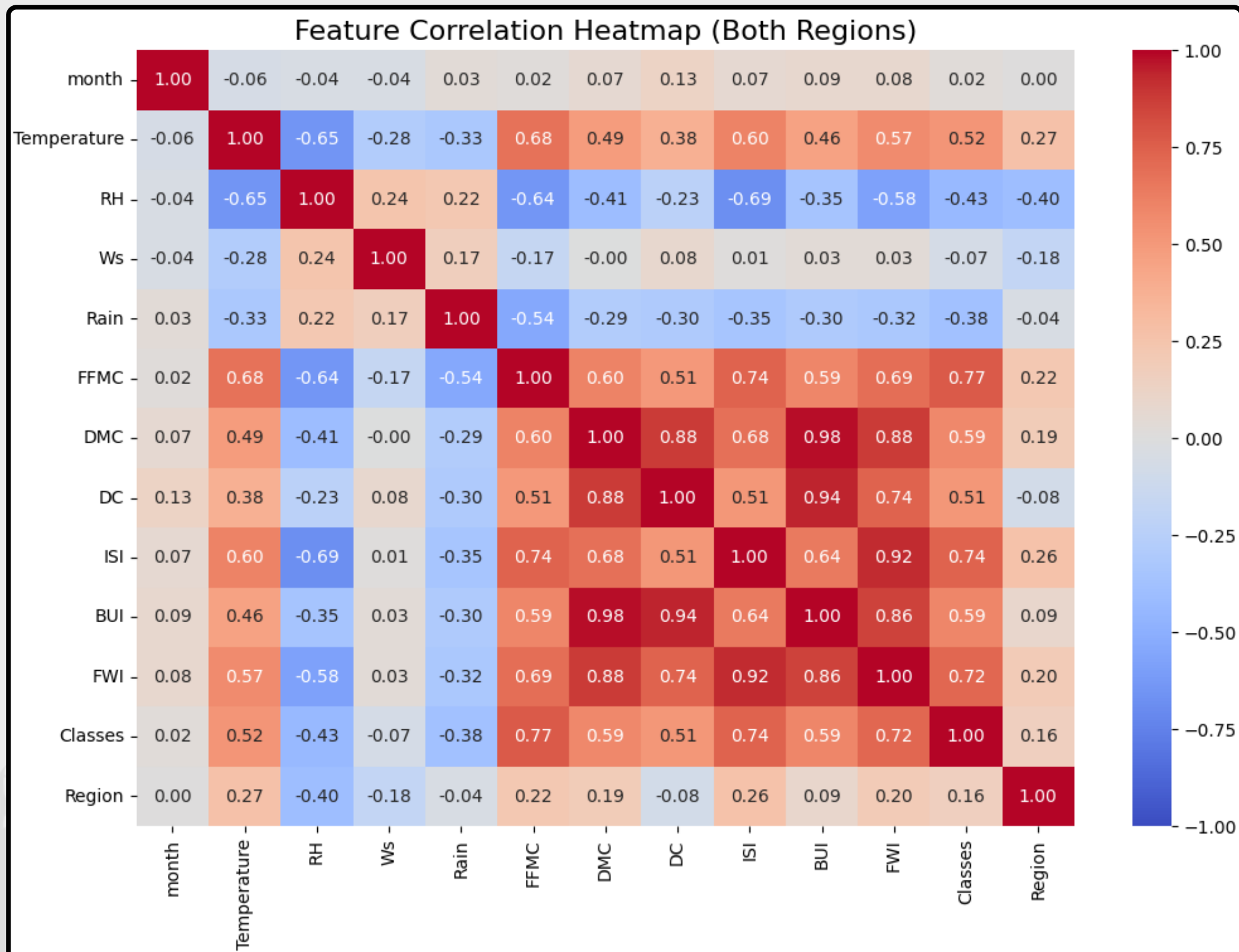
Multicollinearity & Normality Check



- Q-Q plot shows clear deviation from the reference line, indicating violation of multivariate normality
- VIF analysis indicates severe multicollinearity ($VIF > 10$) across most features
- Fire indices (BUI, DMC, FFMC, DC) are highly correlated and redundant
- Rain shows low VIF (~ 1.6), acting as an independent feature
- This multicollinearity affects model stability and motivates dimensionality reduction

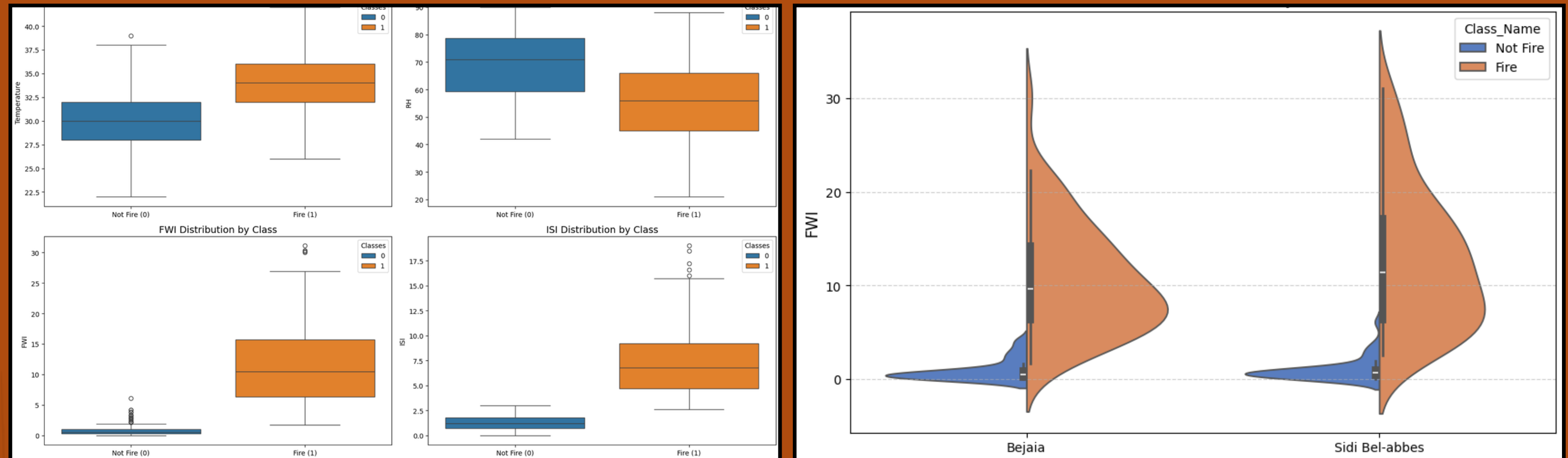


EDA – CORRELATION HEATMAP



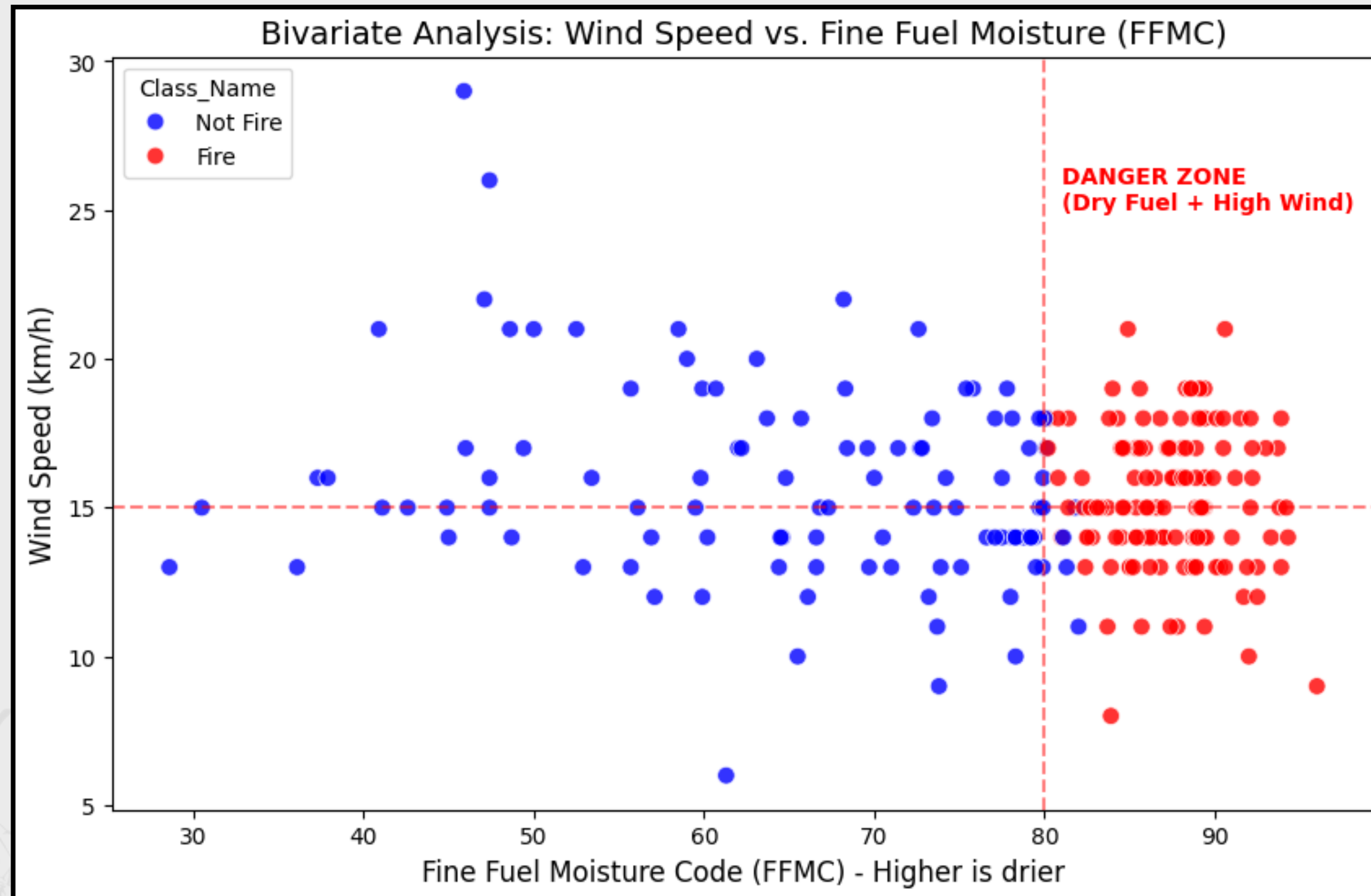
- **Strong Predictors:**
 - FWI, ISI, FPMC show highest positive correlation
 - Temperature shows moderate influence
- **Negative Factors:**
 - RH shows strong negative correlation
 - Rain shows weaker negative impact
- **Multicollinearity:**
 - High correlation among indices (BUI, DMC, FWI, ISI)
 - Indicates redundancy in features
- **Insight:**
 - Multicollinearity affects model stability
 - Dimensionality reduction is required

EDA – CLASS SEPARATION



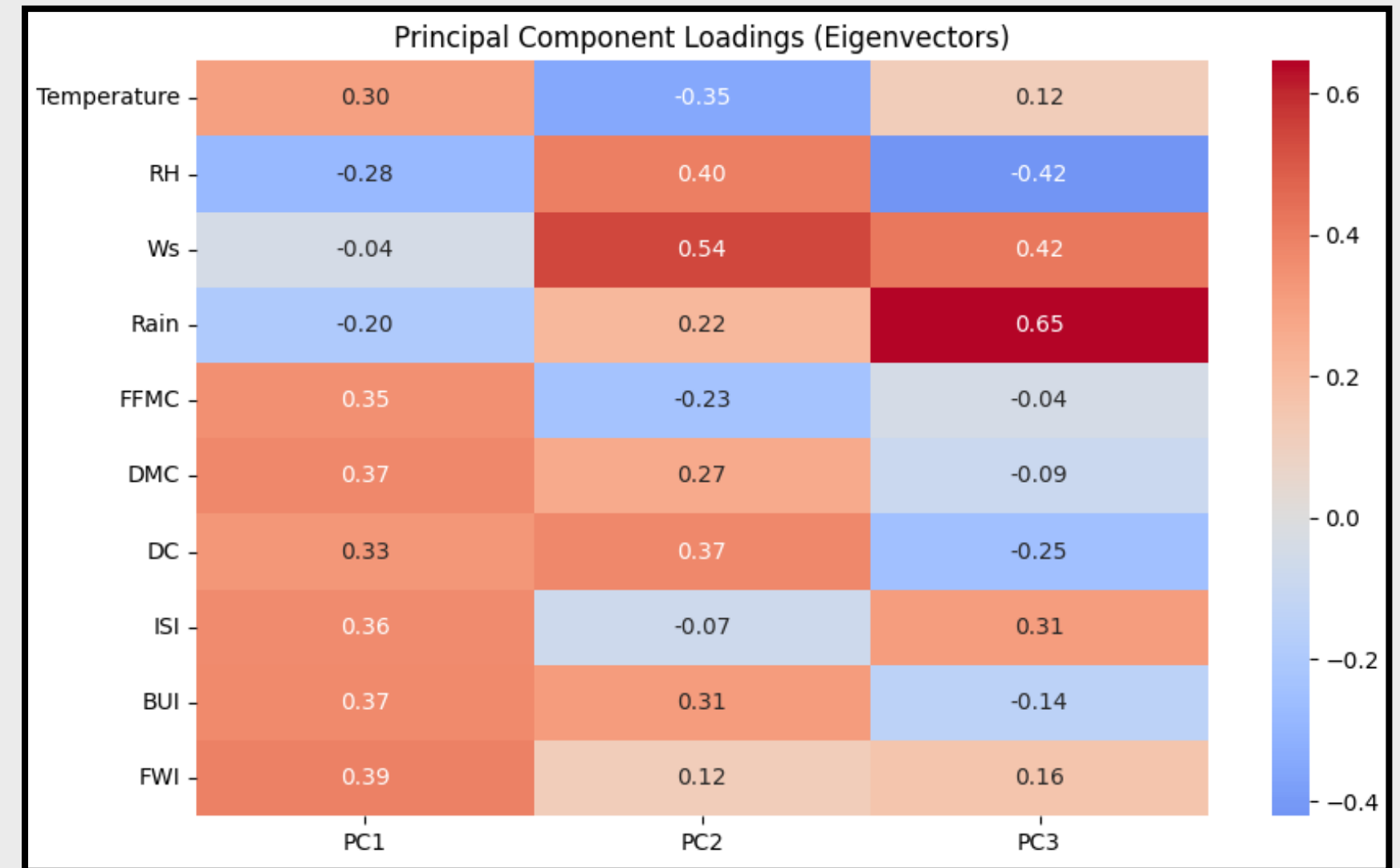
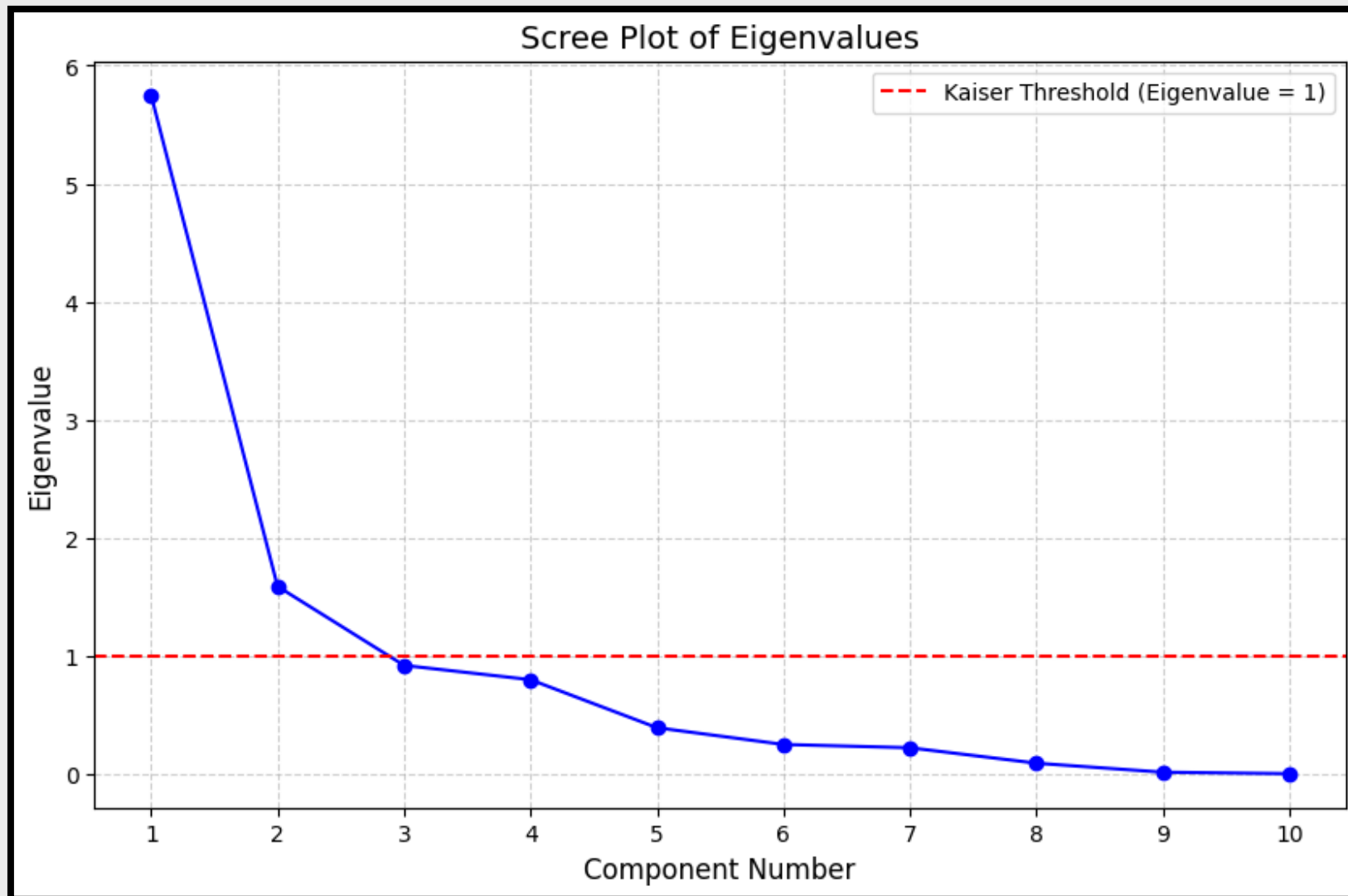
- FWI and ISI show clear separation between fire and non-fire classes
- Fire cases have significantly higher median values compared to non-fire cases
- Minimal overlap in interquartile ranges indicates strong predictive power
- Temperature and RH show wider overlap, making them weaker standalone predictors
- Fire class shows higher density at elevated FWI values across regions
- Sidi Bel-Abbes exhibits wider spread, indicating higher variability in fire conditions
- Non-fire cases are concentrated at lower FWI ranges with less variation
- Distribution patterns confirm that higher dryness strongly increases fire likelihood

EDA – FEATURE INTERACTION



- Wind Speed alone shows weak influence and does not clearly separate fire and non-fire cases
- A clear pattern appears when combined with FFMC, where high dryness (FFMC > 80) with higher wind defines a “Danger Zone” for fire occurrence

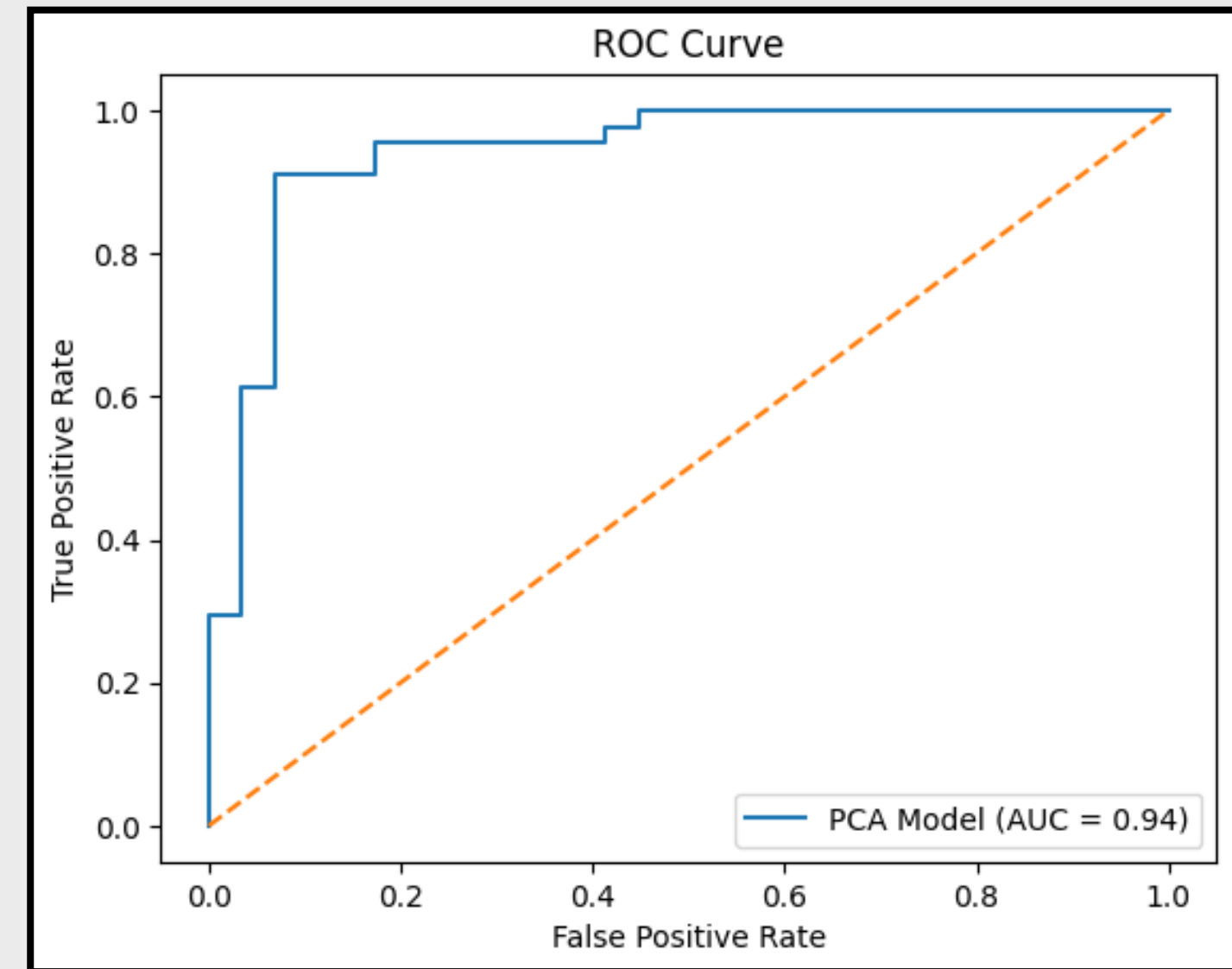
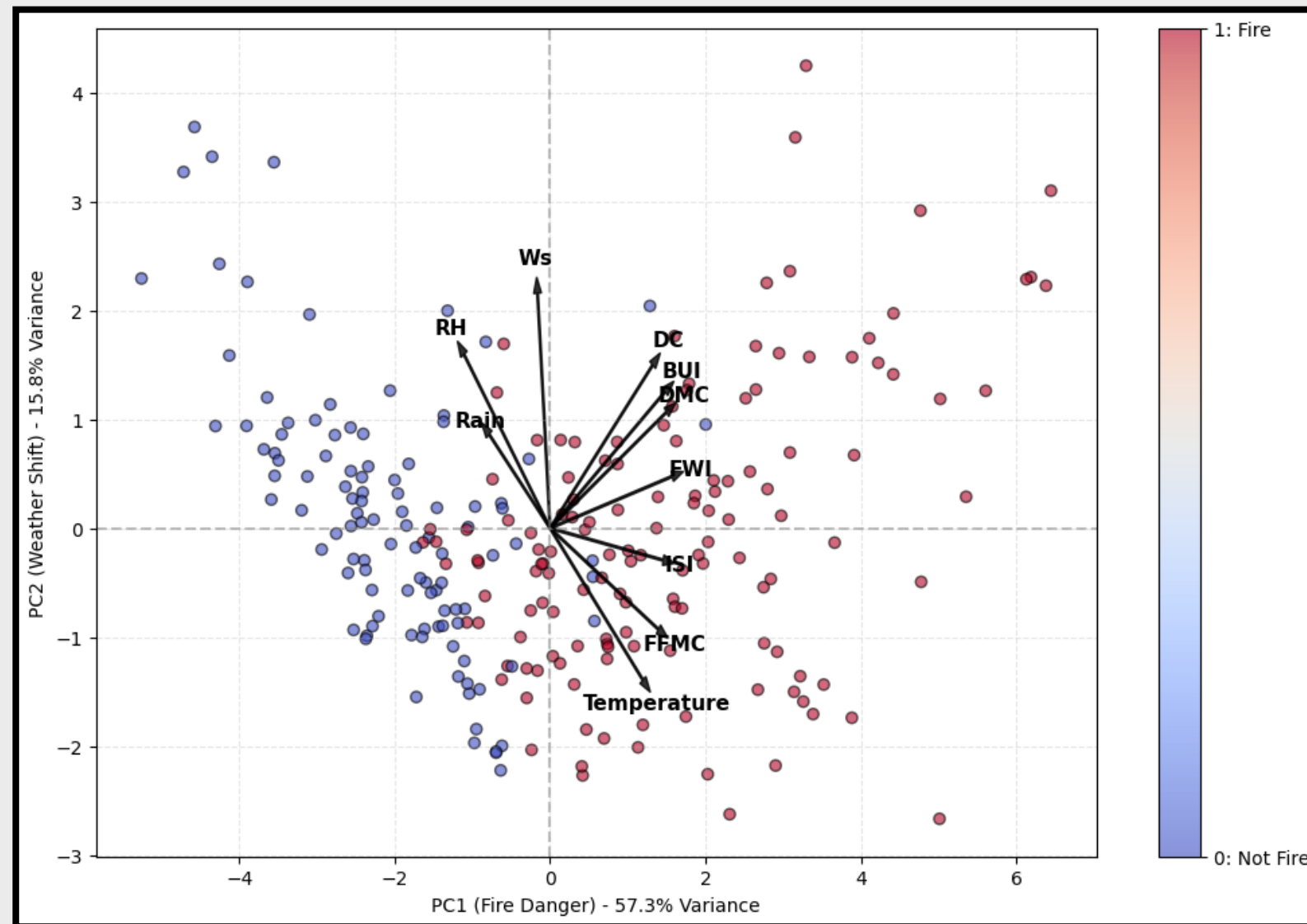
PCA Interpretation



- **PC1 represents the Fire Danger or Dryness factor**, showing strong positive influence from FWI indices and Temperature, along with a negative relationship with Relative Humidity, indicating hot and dry conditions that favor fire occurrence
- **PC2 captures Weather Variability**, driven by Wind Speed, Humidity, and Drought Code, representing dynamic atmospheric changes rather than direct fire risk
- **PC3 reflects the Rain vs Fire Spread trade-off**, where Rain and Wind contribute positively while Humidity shows a negative effect, indicating the balance between rainfall suppressing fires and wind aiding their spread

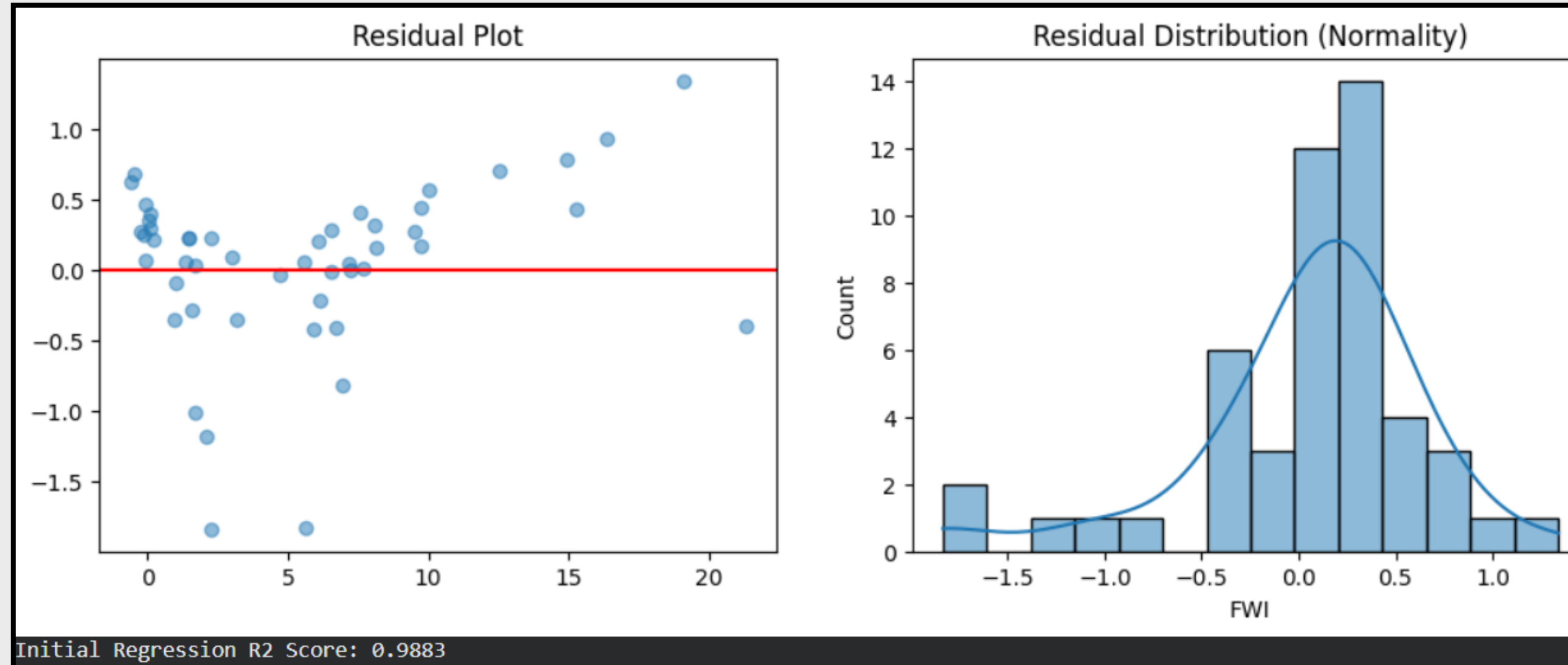


PCA Interpretation



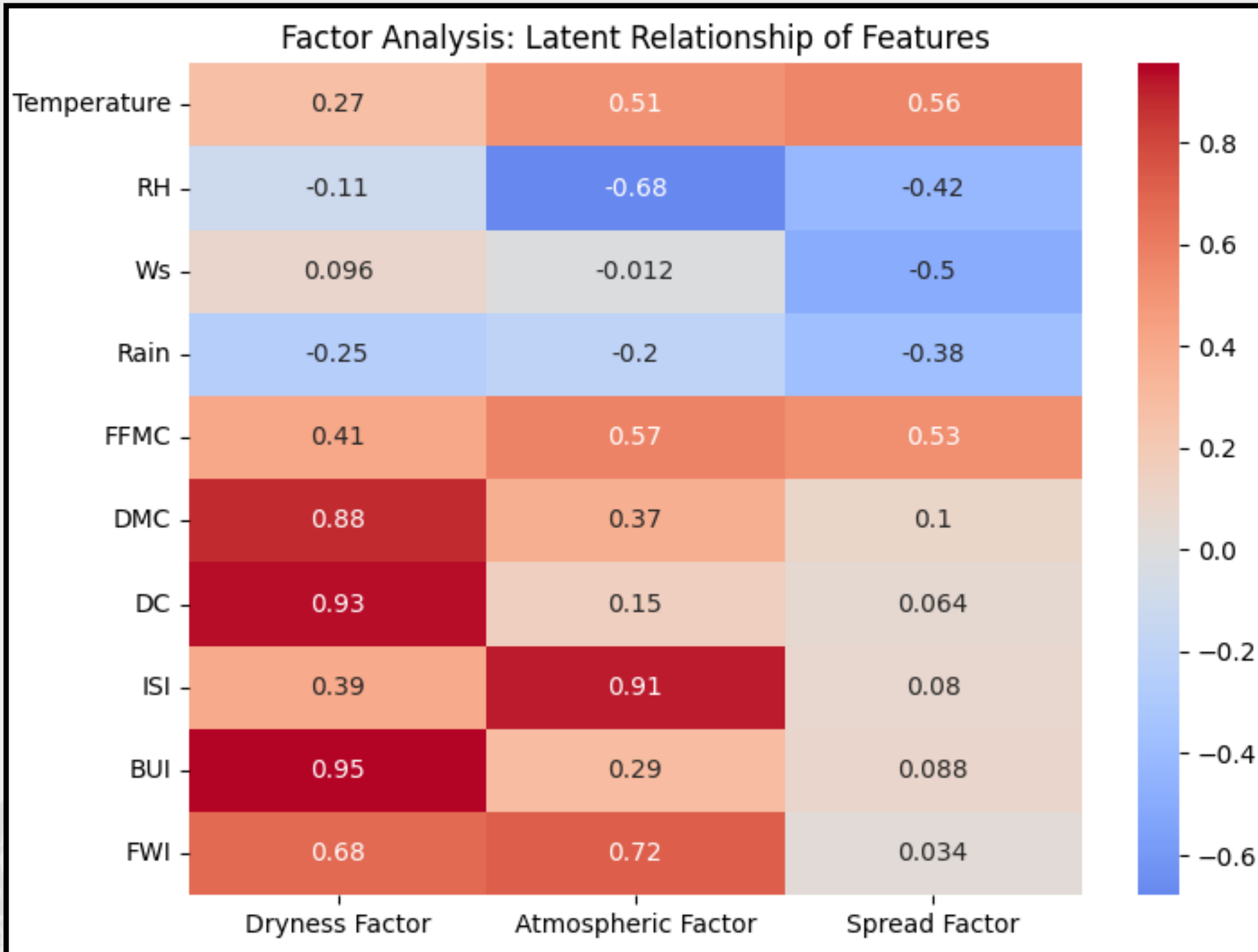
- **PC1 clearly separates fire and non-fire cases, with dryness-related indices (FWI, BUI, DMC, ISI) driving fire occurrence, while RH and Rain reduce fire risk**
- **PC2 captures weather variation but contributes less compared to PC1, showing dryness is the dominant factor**
- **PCA model achieves ~89% accuracy with good stability, and ROC AUC ≈ 0.94 indicates strong classification performance, supported by cross-validation (Avg Accuracy: 0.8805, Std Dev: 0.0664, Interval: 0.8141–0.9469)**

Residual Analysis



- Residuals are mostly centered around zero, indicating a good model fit
- Slight pattern and uneven spread suggest minor non-linearity or heteroscedasticity
- Residuals are approximately normally distributed, with slight skewness
- Model shows very high R^2 (~ 0.99), indicating strong predictive performance
- High accuracy is influenced by multicollinearity and derived nature of FWI

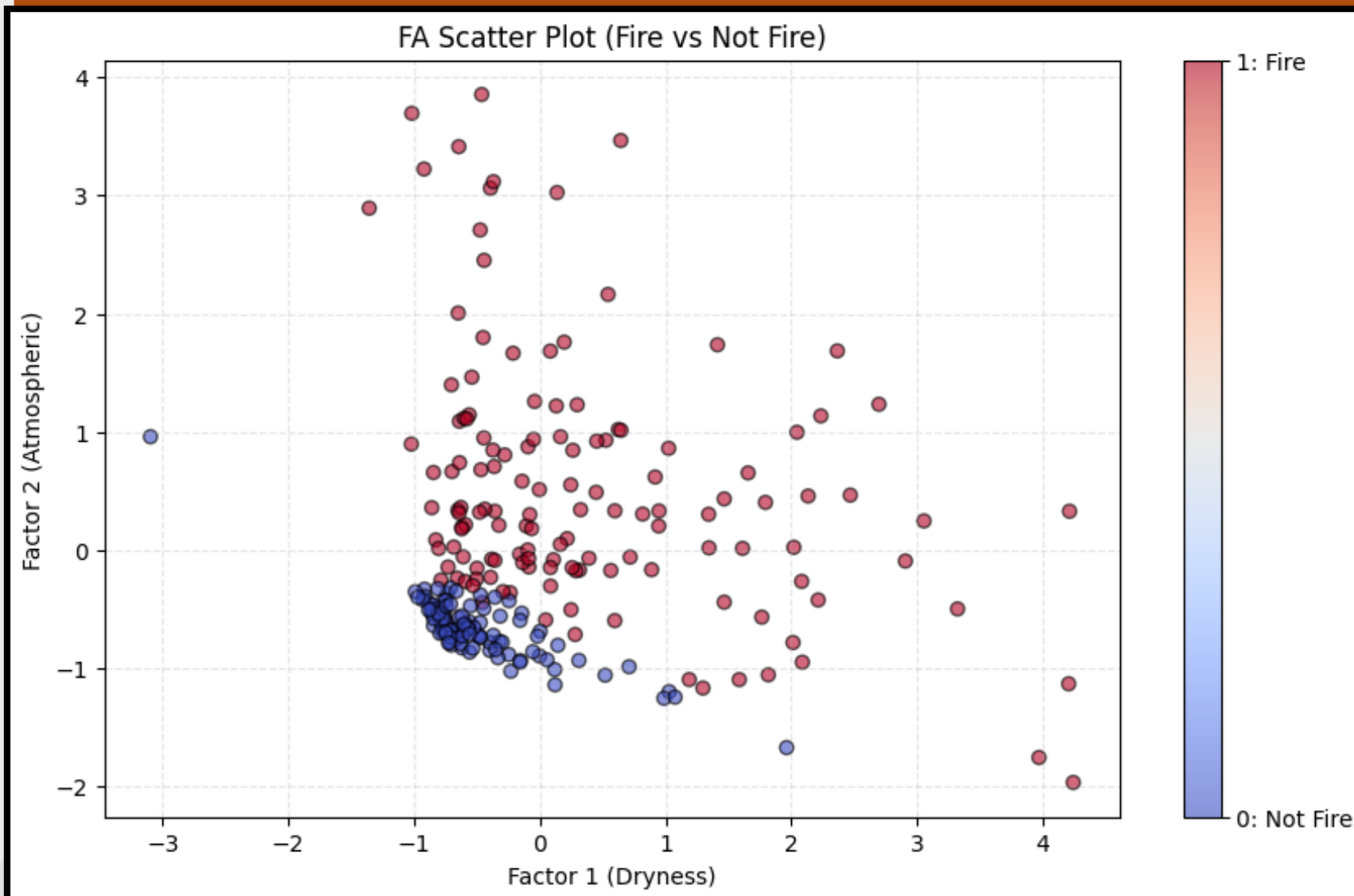
Factor Analysis



- **Dryness Factor:**
 - Driven by BUI, DC, and DMC
 - Represents long-term fuel dryness and fire susceptibility
- **Atmospheric Factor:**
 - Influenced by ISI, Temperature, FFMC and negatively by RH
 - Captures effect of heat and low humidity on fire ignition
- **Spread Factor:**
 - Associated with Temperature, Wind, FFMC, and Rain
 - Represents conditions affecting fire spread



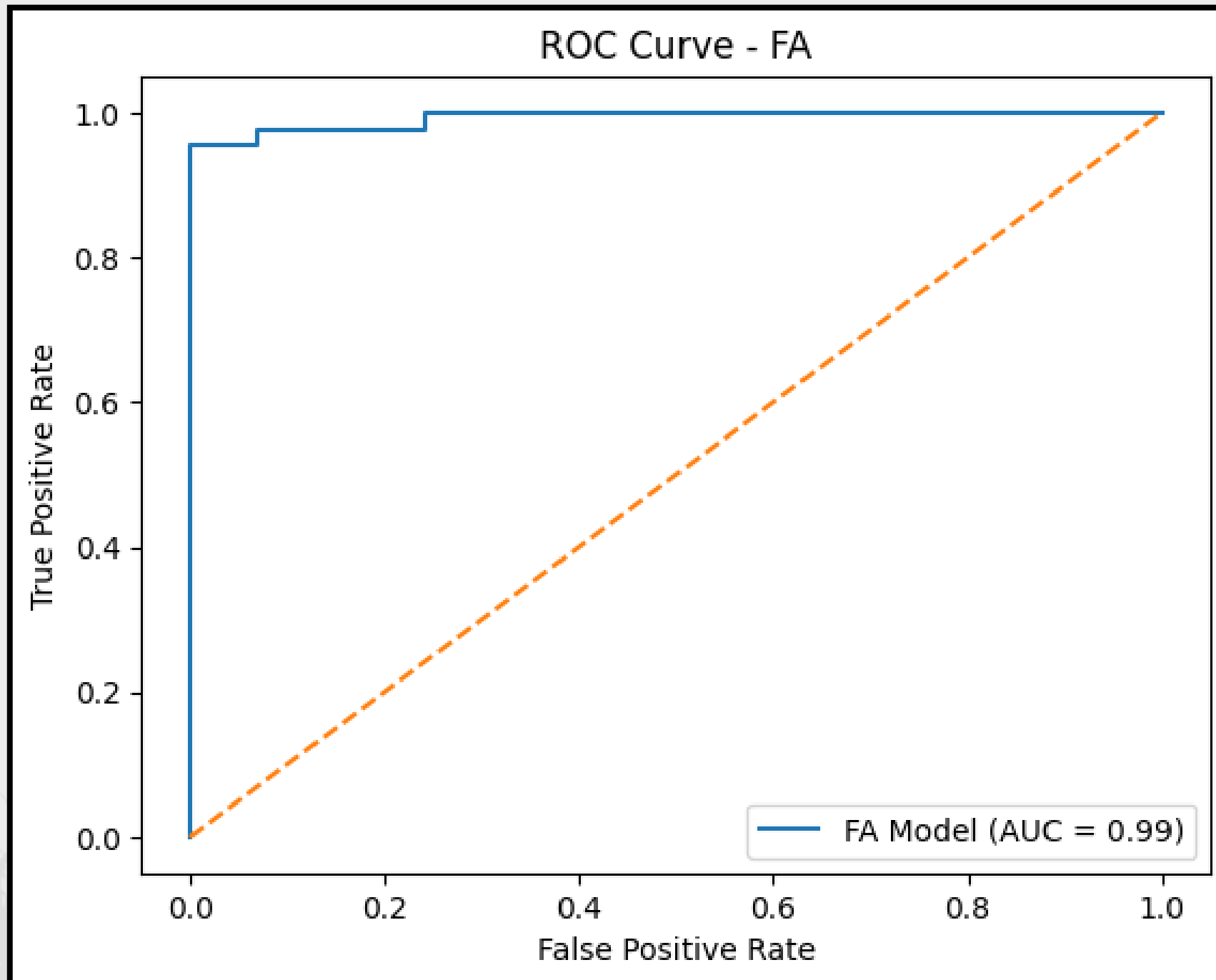
Factor Analysis – Scatter Plot



- **Factor 1 (Dryness) clearly separates fire and non-fire cases**
 - Fire cases occur at higher dryness values
 - Non-fire cases are concentrated in lower dryness region
- **Shows that dryness is the dominant factor for fire occurrence**
- **FA model achieves high accuracy (~97.26%) with only 3 factors**
- **Model is efficient with low training time (~0.025 sec)**



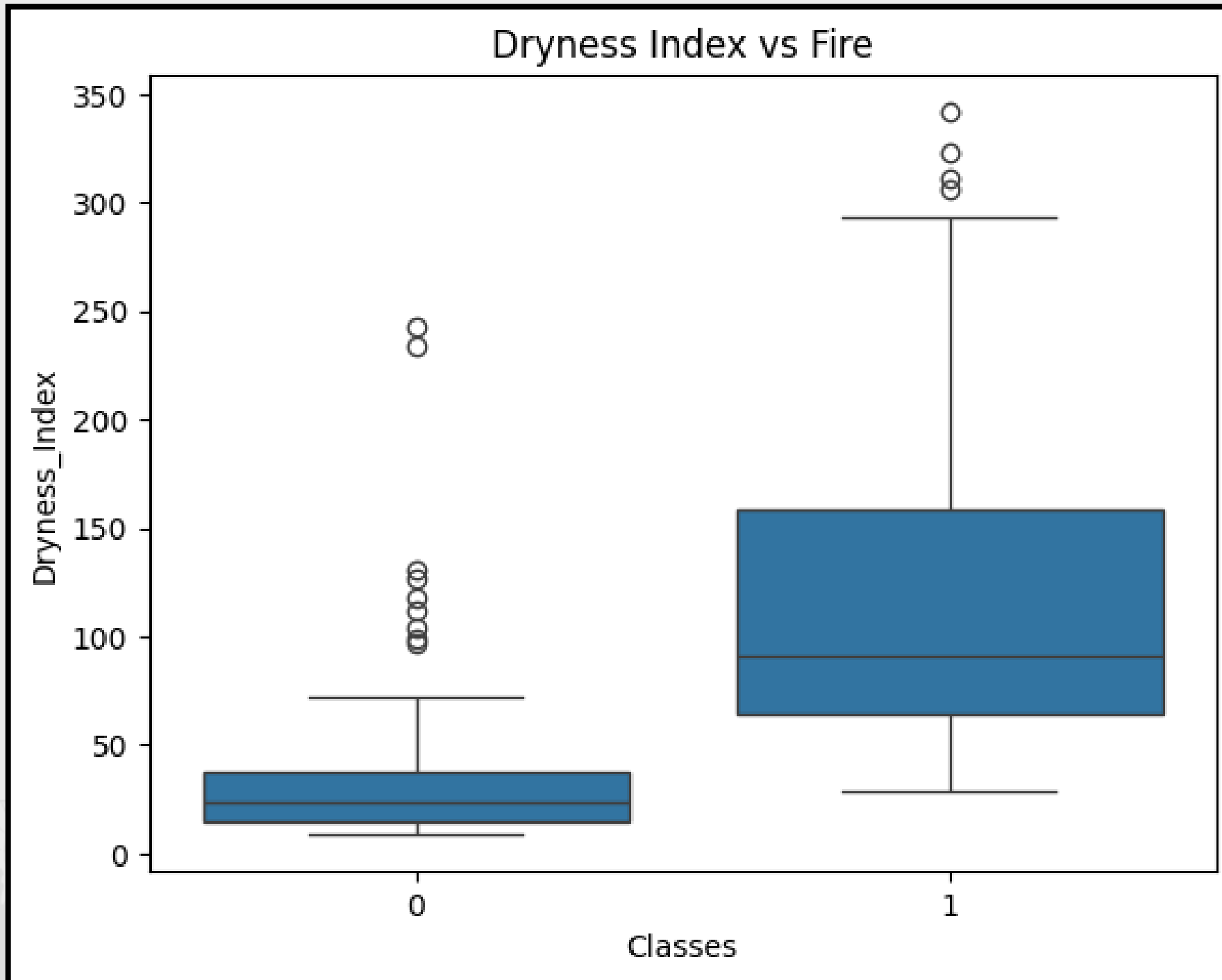
FA Model – Performance



- ROC curve shows $AUC \approx 0.99$, indicating excellent classification performance
- Curve lies near top-left corner $\rightarrow$ high true positive rate, low false positives
- Cross-validation confirms stability
 - Avg Accuracy ≈ 0.95 , low variation
- Precision and recall values are high, showing balanced performance



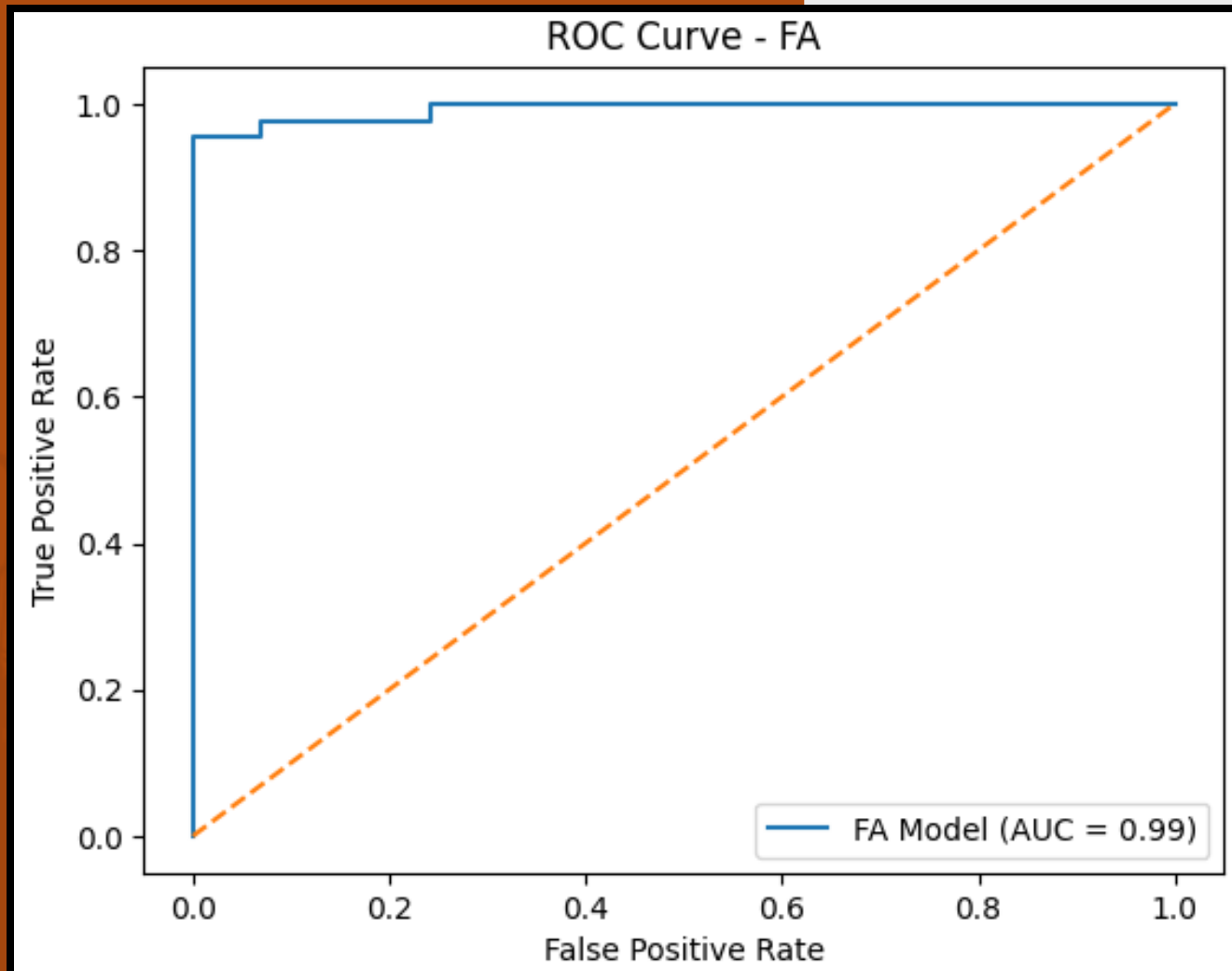
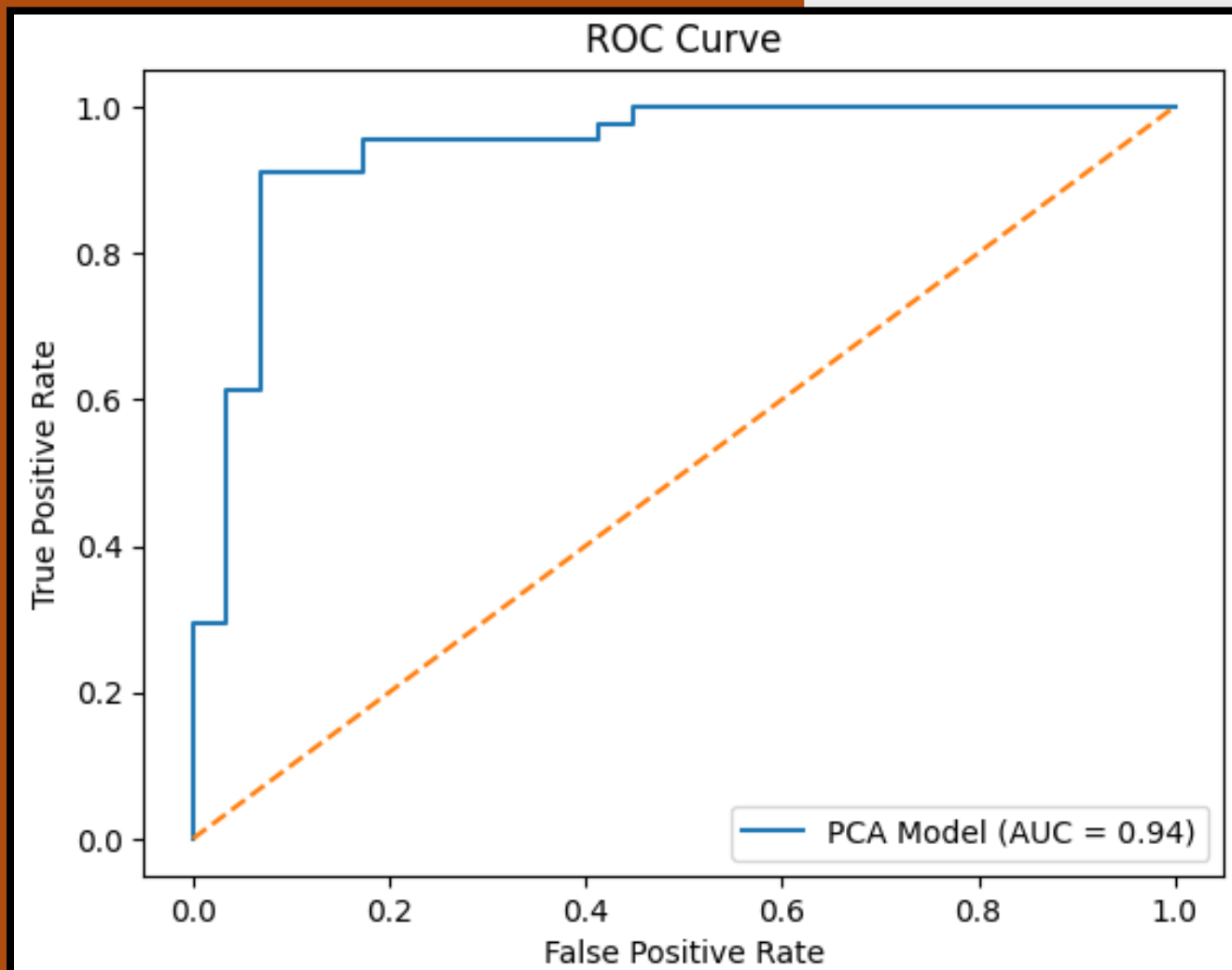
Dryness Index & Model Comparison



- Dryness Index achieved significantly higher values for fire cases, showing clear separation from non-fire cases
- Achieved confirmation that cumulative dryness (BUI, DMC, DC) is a strong driver of fire occurrence
- Rule-based model achieved moderate accuracy (~68%), capturing basic fire patterns
- Machine Learning model achieved high accuracy (~97%), capturing complex feature relationships
- Achieved clear evidence that ML models outperform rule-based approaches in prediction accuracy



MODEL PERFORMANCE



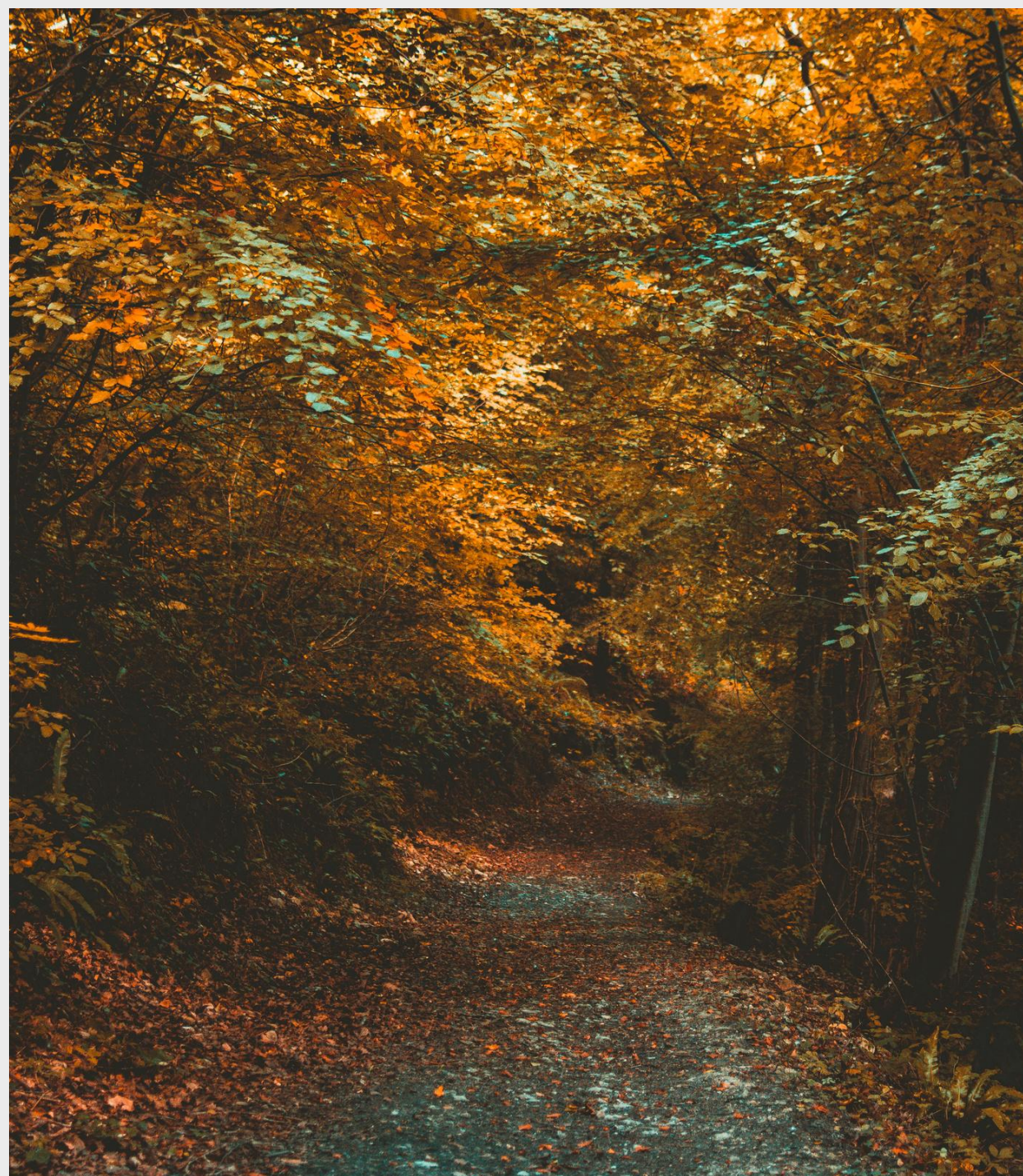
- The performance of models is compared as follows:
 - 1) Raw Model: Accuracy 97.26%, AUC ~0.99
 - 2) PCA Model: Accuracy 89.04%, AUC 0.94
 - 3) FA Model: Accuracy 97.26%, AUC 0.99
- Factor Analysis provides the best balance between accuracy and interpretability
- PCA reduces complexity but slightly lowers performance
- Raw model shows instability due to multicollinearity





Conclusion

- **Factor Analysis is the most reliable and interpretable approach**
- **Fire occurrence is mainly influenced by:**
 - 1) Dryness**
 - 2) Atmospheric conditions**
 - 3) Fire spread factors**
- **Rain and humidity significantly reduce fire risk**
- **The model can be applied for real-time fire prediction systems**



Thank You

